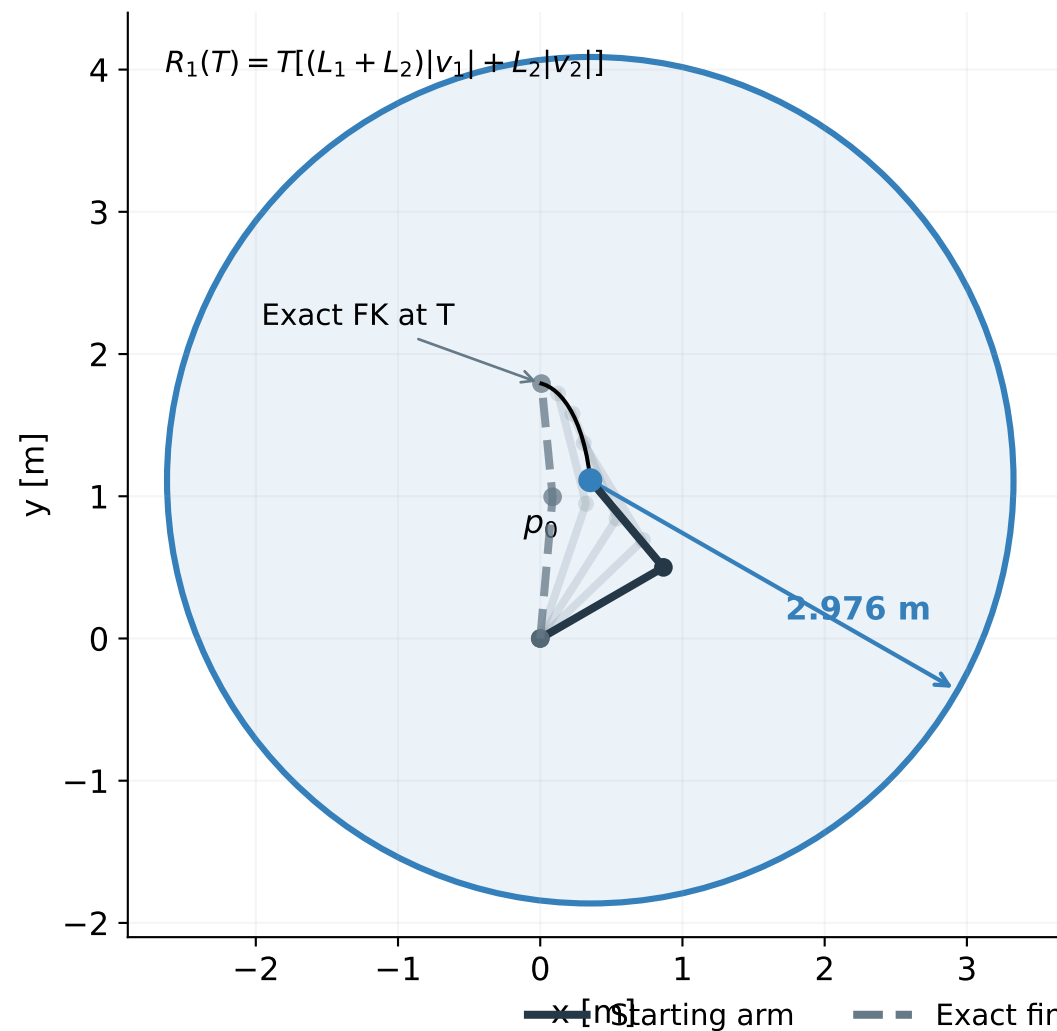
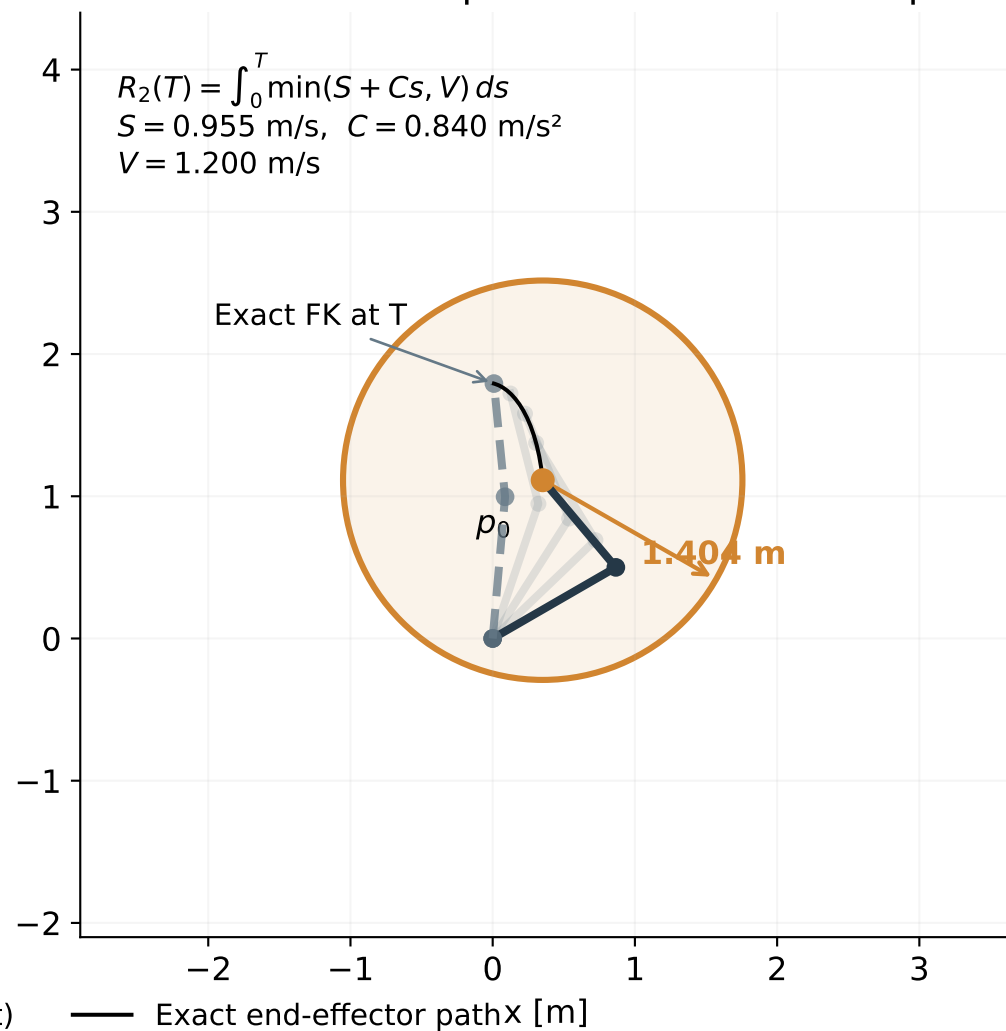


Same held motion, same final arm — two end-effector bounds

L1: worst-case lever arms



Level 2: initial speed + acceleration cap



$T = 1.2 \text{ s}$ • $q(t) = (30^\circ, 100^\circ) + t(0.8, -1.3) \text{ rad/s}$ • Links: 1.0 m, 0.8 m

Shaded disks enclose end-effector displacement for the whole hold; they are not reachable arm configurations.